SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1508

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I Bhuvaneshwari j(192572141)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____________________

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Seasonal Sales – Auto Scaling and Load Balancing

a) Auto Scaling and Load Balancing

A company experiences heavy traffic during seasonal sales such as Black Friday or Diwali. Cloud platforms use **Auto Scaling** to automatically increase or decrease the number of servers based on demand. **Load Balancers** distribute incoming user requests evenly across multiple servers.

- Auto Scaling adds servers during high traffic.
- Removes extra servers during low traffic.
- Load Balancer prevents any single server from becoming overloaded.

b) Reducing Downtime and Improving User Experience

- Prevents server crashes during peak traffic.
- Reduces website response time.
- Ensures fast page loading.
- Provides uninterrupted shopping experience.

c) Role of Monitoring Tools

Monitoring tools continuously track:

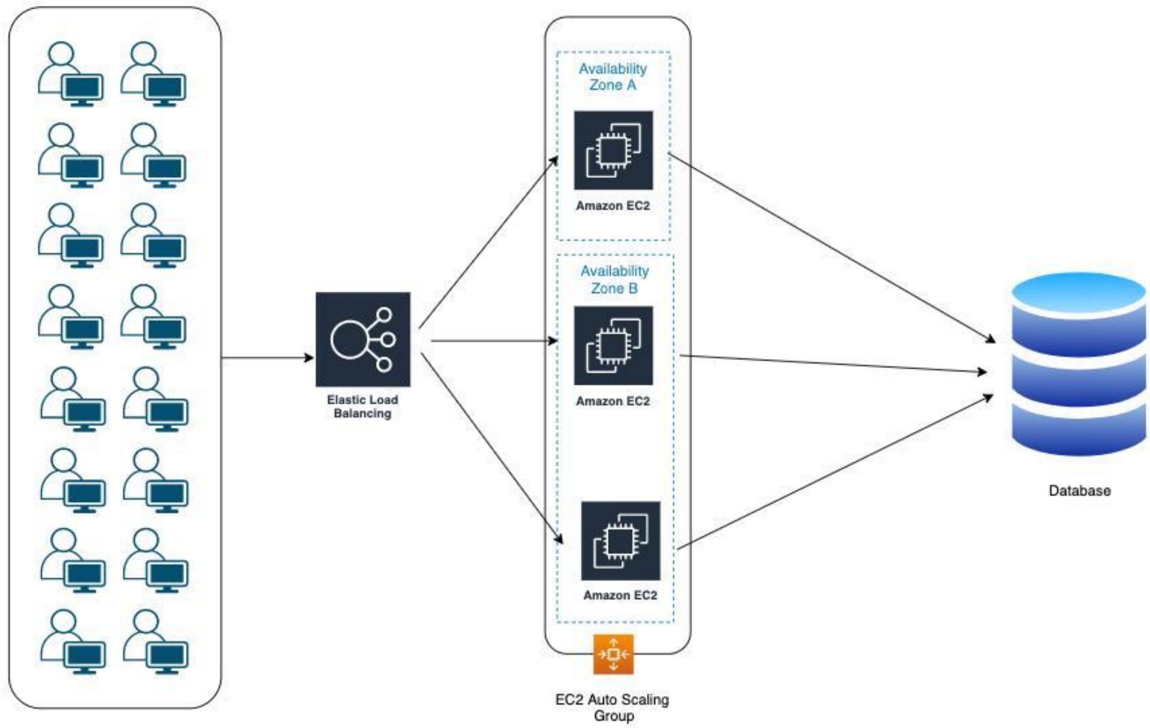
- CPU usage
- Memory usage
- Network traffic
- Response time
- Server health

If resource usage exceeds a threshold, monitoring automatically triggers Auto Scaling.

d) Real-World Example

An online shopping website like **Amazon** experiences millions of users during major sales. Auto Scaling adds more servers, while Load Balancers distribute customer requests. This ensures customers can browse and purchase products without delays.

Cloud Auto Scaling and Load Balancing Architecture



2. Secure Cloud Backup Storage

a) Durability and Data Redundancy

Cloud storage stores multiple copies of data across different servers or regions.

- Automatic replication
- Multiple data copies
- High durability
- Protection against hardware failures

b) Encryption and Access Control

Encryption

- Encrypts data before storage.
- Encrypts data during transmission using SSL/TLS.
- Prevents unauthorized access.

Access Control

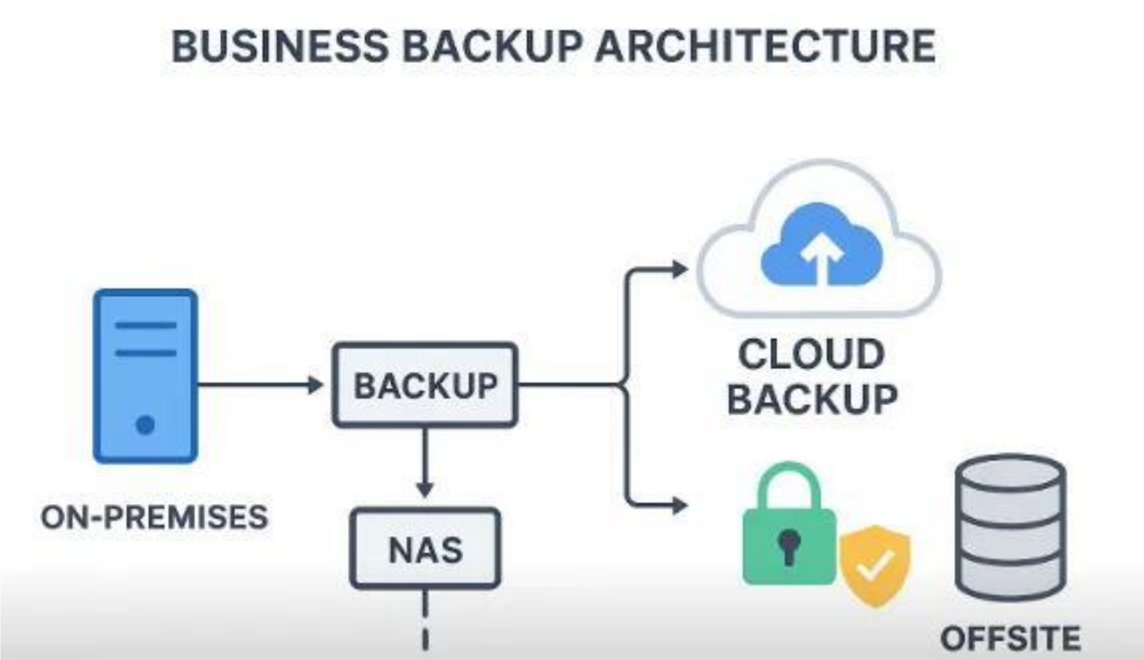
- Role-Based Access Control (RBAC)
- Multi-Factor Authentication (MFA)
- Identity and Access Management (IAM)

c) Cost Advantages
Compared to traditional storage:

- No hardware purchase
- No maintenance cost
- Pay only for storage used
- Easy scalability
- Lower backup infrastructure cost

d) Disaster Recovery Example
A company's database is accidentally deleted. Cloud backup restores the latest backup from another region, allowing the business to resume operations with minimal downtime.

Cloud Backup and Disaster Recovery Architecture



3. Global Mobile App Deployment

a) Content Delivery Networks (CDNs)

A **Content Delivery Network (CDN)** stores cached content on servers located close to users.
Benefits:

- Faster content delivery
- Reduced latency
- Lower bandwidth usage
- Improved application performance

b) Edge Servers

Edge servers are located near users.
They:

- Deliver content quickly.
- Reduce the distance data travels.
- Improve page loading speed.
- Reduce network congestion.

c) Global Availability

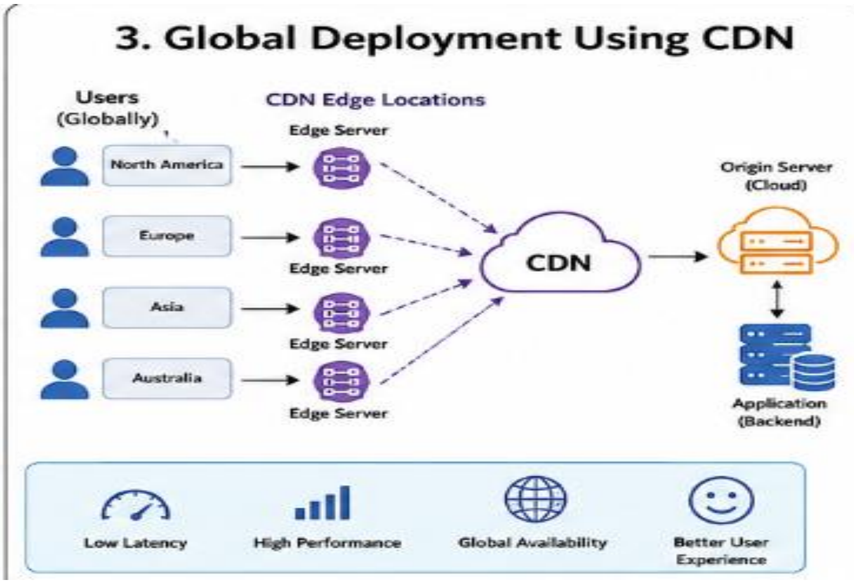
Cloud providers offer:

- Multiple data centers worldwide
- Global load balancing
- Regional replication
- High availability across continents

d) Real-Time Example

Streaming services like **Netflix** or online games use CDNs and edge servers to stream videos and game content with minimal buffering and low latency for users worldwide.

Global Depolyment Using CDN



4. Financial Institution – Cloud Security and Compliance

a) Regulatory Compliance and Auditing

Cloud providers support compliance standards such as:

- ISO 27001
- PCI DSS
- GDPR
- SOC 2

They provide audit logs to track all user activities.

b) Logging, Monitoring, and Encryption

Logging

- Records every user activity.
- Helps during audits.
- Detects suspicious activities.

Monitoring

- Tracks system performance.
- Detects security threats.
- Generates alerts for unusual behavior.

Encryption

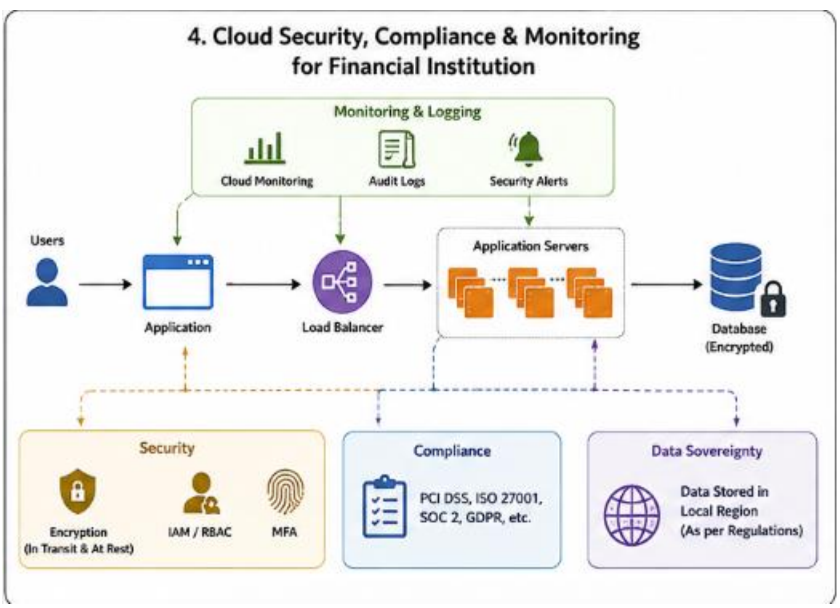
- Encrypts stored data.
- Encrypts transmitted data.
- Protects customer financial information.

c) Data Sovereignty
Organizations can:

- Store data in approved geographic regions.
- Select specific cloud regions.
- Follow local government regulations.
- Restrict data movement across countries.

d) Example
A bank processes online money transfers. Customer account details are encrypted, every transaction is logged, and monitoring systems detect suspicious login attempts. Data remains stored within the required country to meet regulatory requirements.

Cloud Security Compliance and Monitoring For Financial Institution



5. Containerization and Kubernetes

a) Containers vs Virtual Machines

Containers	Virtual Machines
Lightweight	Heavyweight
Share host OS	Separate operating system
Fast startup	Slow startup
Less memory usage	More memory usage
High performance	Lower performance

b) Kubernetes Orchestration

Kubernetes simplifies deployment by:

- Automatically deploying containers.
- Scaling applications based on demand.
- Restarting failed containers.
- Load balancing traffic.
- Managing container networking.

c) Portability

Containers can run consistently across:

- AWS
- Microsoft Azure
- Google Cloud
- On-premises servers

This makes applications highly portable and avoids vendor lock-in.

d) Microservices Example

An **online food delivery application** can be divided into multiple microservices:

- User Service
- Login Service
- Restaurant Service
- Order Service
- Payment Service
- Delivery Tracking Service

Each microservice runs in its own container, allowing independent deployment, scaling, and updates.

Containerization and Kubernetes Orchestration